Squadron Koller: Team and Roles (Mission 05)

Mission: 05 | Dual-Path GraphRAG, Hybrid Search & The Google NotebookLM Grounding Lab

Squadron: Koller | **Officers:** [Antoun Al Salameh 5019770] [Ragni Chawala 5012421]

Waypoint: Crab Pulsar Remnant / Messier 1 (SEC-05-M01) | **Subsystem:** 05 - Dual-Path Associative Memory Core

Course: Conversational AI - Winter Semester 2026/27 | **Station(s):** All

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Mission: 05 Dual-Path GraphRAG, Hybrid Search & the Google NotebookLM Grounding Lab

Repository: https://github.com/Unisvet/couse-c-ai/tree/main/weeks/week-05

1. Team Roster and Station Assignments

The following table outlines the operational roles for Mission 05 (assignments carried over from Mission 02; edit if your squadron redistributed the work).

Name	Squadron Station	Mission 05 Role
[Antoun Al Salameh 5019770]	Data & Payload Specialist	Part 3: NotebookLM Grounding & Citation Audit
[Antoun Al Salameh 5019770]	Communications & Protocol Officer	Part 2: BM25 & Reciprocal Rank Fusion
[Ragni Chawala 5012421]	Security & Ethics Sentinel	Part 4: Injection Defense & RAG Triad Audit
[Ragni Chawala 5012421]	Systems Architect	Part 1: HNSW Vector Index
[Ragni Chawala 5012421]	AI Security Officer	Mission 05 EU AI Act Compliance

2. Detailed Role Responsibilities & Work Distribution

Systems Architect

Lead engineer for the HNSW multi-layer vector proximity index (Malkov & Yashunin, 2018). Implements the cosine core on unit vectors, exponential-decay layer assignment (l = floor(-ln U · mL), mL = 1/ln M), greedy search_layer() traversal, and a latency benchmark of HNSW against flat linear search at N = 25,000 768-dimensional vectors. Also owns the integrated pipeline and the repository layout under weeks/week-05/.

Communications & Protocol Officer

Implements the Okapi BM25 scoring engine (k1 = 1.5, b = 0.75) and the Reciprocal Rank Fusion combiner (k = 60), and demonstrates that the hybrid pipeline retrieves both conceptual queries (dense-dominated) and exact serial codes (BM25-dominated).

Data & Payload Specialist

Runs the live experimental audit of Google NotebookLM: ingests Documents A, B and C, formulates 5 cross-document queries and computes Citation Precision, runs 3 negative-constraint probes for epistemic refusal, and generates and scores an Audio Overview excerpt for Factual Fidelity, Persona Differentiation and Pedagogical Value. Produces notebooklm_audit_report.md from real observations.

Security & Ethics Sentinel

Builds the Indirect Prompt Injection detector (heuristic plus embedding layers), the automated RAG Triad scorer (Context Relevance, Faithfulness, Answer Relevance) and the structured JSON provenance audit log recording query, retrieved chunk IDs, similarity scores, injection scan result and final answer.

AI Security Officer (Compliance)

Evaluates Subsystem 05 against the EU AI Act (Regulation 2024/1689): prohibited practices, risk classification, Article 50 transparency, and data-protection hazards of the provenance log and third-party NotebookLM use. Delivers the compliance review and the action list before deployment.

3. Submission Structure

File (weeks/week-05/)	Owner
01_HNSW_Vector_Index.py	Systems Architect
02_BM25_and_RRF_Combiner.py	Communications & Protocol Officer
03_NotebookLM_RAG_Audit.py, notebooklm_audit_report.md	Data & Payload Specialist
04_RAG_Provenance_and_Safety.py	Security & Ethics Sentinel